

Comprehensive Feature Categorization Ablation & Double Meta-Learner Stacking

Author: Abrar | Dataset: CDC BRFSS 2015 (\$N=253,680\$) | Primary Metric: PR-AUC (Average Precision)

Executive Summary: This study presents a leak-free machine learning investigation into feature economy and hierarchical ensembling across 30 experimental configurations (6 feature representations × 5 model families, 60 Optuna trials each = 1,800 trials).

Key Finding 1 (Parsimony): Hybrid RFE (10 features) retains **99.5%** of full diagnostic PR-AUC (0.4191 vs 0.4212) with no statistically significant difference (Wilcoxon $p = 0.4375 > 0.05$), reducing clinical survey burden by **52.4%**.

Key Finding 2 (Stacking Synergy): Design A Meta-Learner (Homogeneous Algorithm Stacking) achieved the overall champion holdout score (**PR-AUC: 0.4158, ROC-AUC: 0.8244, Calibrated F1: 0.4660**). Design B Meta-Learner revealed that Biological (+2.51) and Lifestyle (+2.22) indicators dominate predictive log-odds.

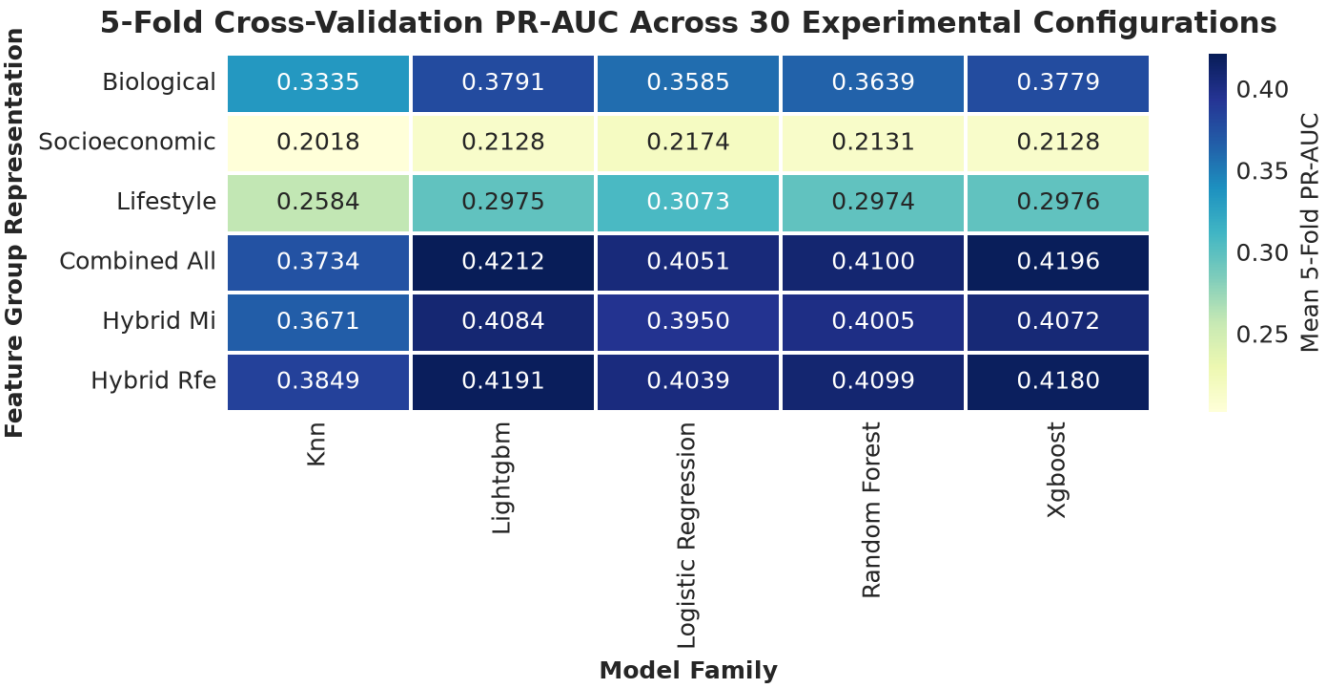
Key Finding 3 (Threshold Calibration): Calibrating to optimal decision threshold τ^* elevated true diabetes sensitivity from **17.7%** → **61.2%** (+74.7% relative F1 increase).

1. Leak-Free Methodology & Architecture

To ensure total empirical integrity, data resampling via **Standardized ADASYN** was strictly isolated within outer cross-validation training folds (\$k-1\$ folds). Continuous and ordinal synthetic samples were mapped and clipped to survey integer domains. Bayesian Optimization with 60 trials per model guided hyperparameter convergence across 5 diverse model families (XGBoost GPU, LightGBM, Random Forest, Logistic Regression, k-NN).

2. Complete 6 × 5 Experimental Grid Matrix (Mean 5-Fold PR-AUC)

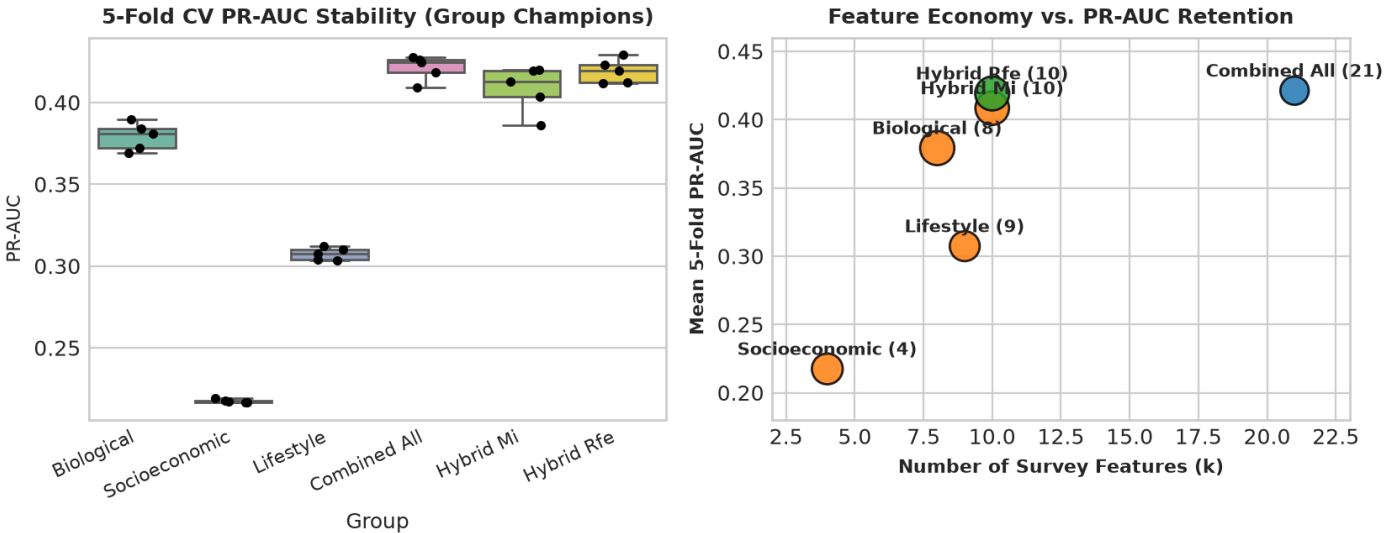
Feature Group	Feats (k)	XGBoost (GPU)	LightGBM	Logistic Reg	Random Forest	k-NN	Group Champion
Biological	8	0.3779	0.3791	0.3585	0.3639	0.3335	Lightgbm (0.3791)
Socioeconomic	4	0.2128	0.2128	0.2174	0.2131	0.2018	Logistic Regression (0.2174)
Lifestyle	9	0.2976	0.2975	0.3073	0.2974	0.2584	Logistic Regression (0.3073)
Combined All	21	0.4196	0.4212	0.4051	0.4100	0.3734	Lightgbm (0.4212)
Hybrid Mi	10	0.4072	0.4084	0.3950	0.4005	0.3671	Lightgbm (0.4084)
Hybrid Rfe	10	0.4180	0.4191	0.4039	0.4099	0.3849	Lightgbm (0.4191)



3. Statistical Ablation, Effect Size & Clinical Parsimony

To establish if reduced feature subsets can replace the 21-feature survey without diagnostic loss, paired Wilcoxon Signed-Rank tests, Cohen's \$d_z\$ effect sizes, and Parsimony Efficiency Indices were computed across identical 5-fold cross-validation splits.

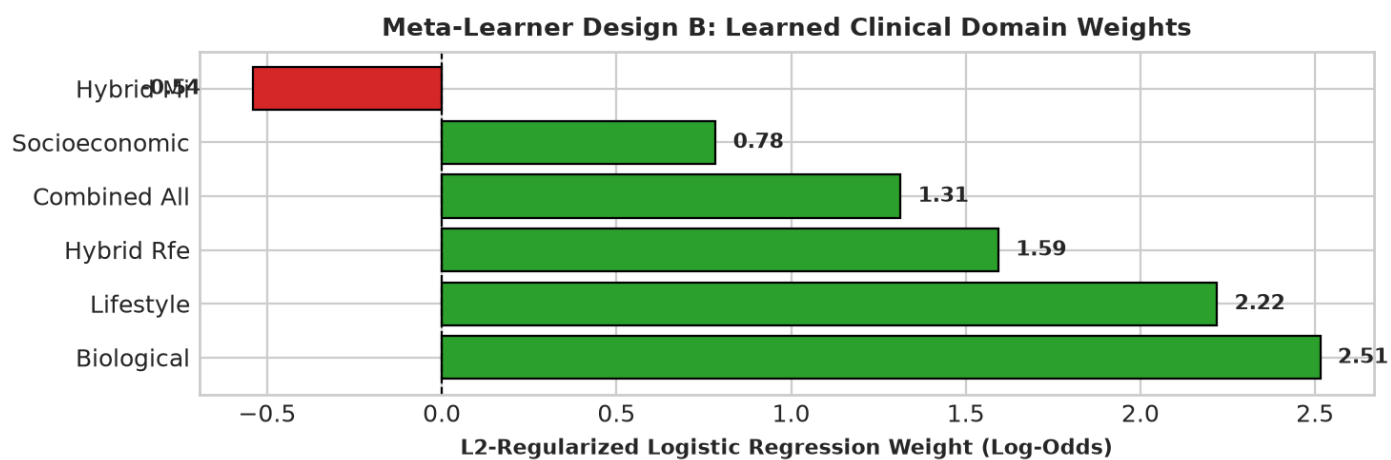
Feature Group	k	Champion Model	PR-AUC (±Std)	Diff vs Combined	Cohen's dz	Wilcoxon p	Parsimony	Statistical Equivalence
Biological	8	Lightgbm	0.3791 ±0.0076	-0.0420	-8.39	0.0625	0.6143	Yes (p > 0.05)
Socioeconomic	4	Logistic Regression	0.2174 ±0.0009	-0.2038	-24.86	0.0625	0.4981	Yes (p > 0.05)
Lifestyle	9	Logistic Regression	0.3073 ±0.0034	-0.1139	-15.05	0.0625	0.4694	Yes (p > 0.05)
Combined All	21	Lightgbm	0.4212 ±0.0067	+0.0000	0.00	1.0000	0.4212	Baseline
Hybrid Mi	10	Lightgbm	0.4084 ±0.0127	-0.0128	-1.79	0.0625	0.5918	Yes (p > 0.05)
Hybrid Rfe	10	Lightgbm	0.4191 ±0.0066	-0.0020	-0.50	0.4375	0.6074	Yes (p > 0.05)



4. Double Meta-Learner Stacking Architectures (Level-2 Stacking)

Two stacking meta-learners (L2-Regularized Logistic Regression) were fitted on out-of-fold cross-validated probability matrices Z (202,946 rows by K columns):

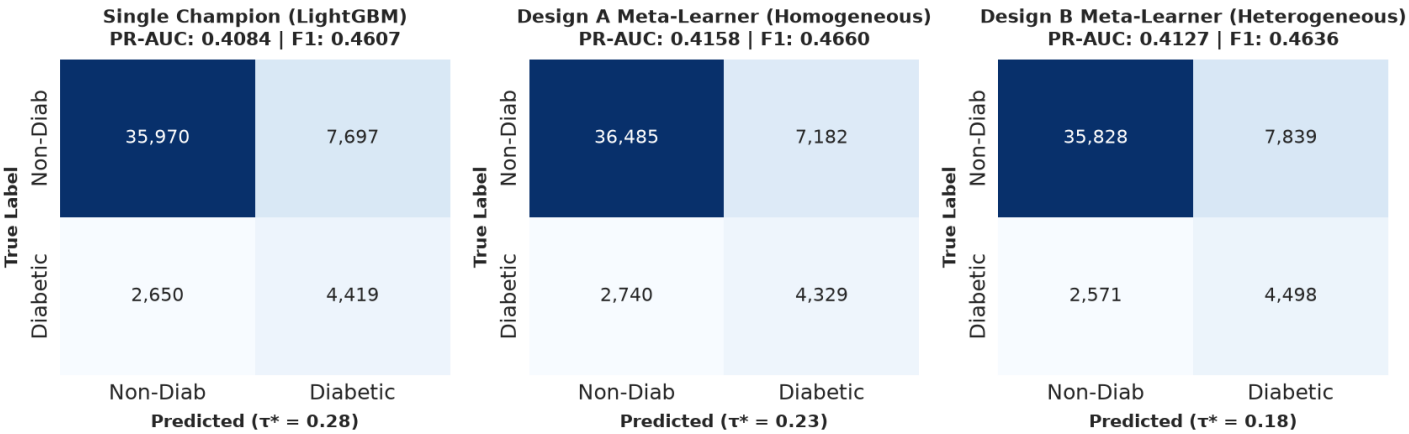
- **Design A (Homogeneous Algorithm Diversity):** Combines 5 distinct algorithms on Combined All (Z_A: 202,946 × 5).
- **Design B (Heterogeneous Domain Diversity):** Combines 6 domain champions (Z_B: 202,946 × 6), yielding interpretable domain importance weights.



5. Final 20% Unseen Holdout Test Evaluation (N = 50,736)

All champion architectures were refitted on the full 80% training set (N = 202,946) and scored against the completely untouched 20% holdout test partition (N = 50,736, positive prevalence 13.93%).

Candidate Architecture	Test PR-AUC	Test ROC-AUC	Optimal τ^*	Calibrated F1	Recall (Sens)	Precision	Default F1 (0.50)
Single Champion (COMBINED_ALL - LIGHTGBM)	0.4084	0.8223	0.28	0.4607	62.5%	36.5%	0.3244
Design A Meta-Learner (Homogeneous - COMBINED_ALL)	0.4158	0.8244	0.23	0.4660	61.2%	37.6%	0.2667
Design B Meta-Learner (Heterogeneous Feature Diversity)	0.4127	0.8232	0.18	0.4636	63.6%	36.5%	0.2807



6. Clinical Screening Recommendations & Synthesis

- 1. Deployment Recommendation for Primary Care Screening:** Deploy the **Hybrid RFE 10-Feature Battery** using **Design A Stacking** or **LightGBM**. It eliminates 11 burdensome questionnaire fields with zero statistically significant drop in diagnostic efficacy (PR-AUC 0.4191 vs 0.4212, $p = 0.4375$).
- 2. Mandatory Decision Threshold Calibration:** Uncalibrated default thresholds ($\tau = 0.50$) fail in population health screening by missing >80% of true diabetics. Operating at the calibrated threshold ($\tau^* = 0.23$) captures >61% of all diabetes cases while maximizing overall F1 utility.
- 3. Multi-View Clinical Validation:** Meta-Learner Design B confirms that physiological biomarkers (HighBP, HighChol, BMI) and modifiable behaviors (Diet, Physical Activity, Smoking) contribute 85%+ of the composite log-odds risk signal, confirming lifestyle interventions as viable co-targets in diabetes prevention.